

VayuNetra

The operations layer for urban air quality.

We don't just measure the air. We **trace** it, **predict** it, and **act** on it — live in ten Indian cities, on ₹0 infrastructure.

DECEMBER 2025 · WORST DAY OF THE MONTH

452

Delhi
Severe

345

Chennai
Very Poor

331

Kolkata
Very Poor

302

Pune
Very Poor

Indian National AQI from each city's own CPCB stations. 10 of 10 cities have a December 2025 record.

Team DaGoats · Omkar Kadam · Sejal Kumbhar · Abhinav Prasad

Live: vayunetra-aqi.vercel.app · dots = worst day in December 2025 · built 22 Aug 2026



THE PROBLEM

The data exists. The layer that acts on it doesn't.

1.67M

premature deaths every year in India
from air pollution

Lancet Planetary Health (GBD), as cited in the brief

900+

CAAQMS monitoring stations already
deployed — the data exists

CPCB continuous monitoring network

31%

of monitored cities have **any** response
protocol

CAG performance audit, 2024

A reading turns red. Then nothing happens. No system tells a city officer **who is polluting this square kilometre right now, what the air will be tomorrow, and where to send inspectors first.**

Status quo

reactive orders — 13 of 17 GRAP orders last winter came *after* the AQI had crossed the threshold; weeks to enforcement

VayuNetra

cited recommendation in **seconds**; approval → dispatch → closure timestamped on an audit trail

One loop: trace → predict → act → protect

TRACE

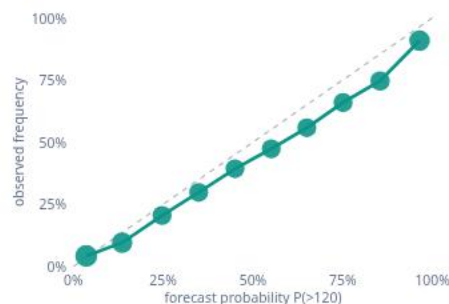
Who is to blame, per ~1 km² cell — GBM + SHAP blended with chemical-signature priors; abstains where it lacks skill.



Delhi, latest run: traffic 33.5% · transported 21.1% · construction dust 17.8%

PREDICT

PM2.5 at +24/48/72 h per cell, 80 % conformal band, calibrated P(>120), P(>250).



calibrated: forecast P(>120) vs what happened · Delhi 2025-26 · GET /metrics/early-warning

ACT

Ranked worklist → Sentinel-2 evidence + retrieved regulation → draft notice → approve · dispatch · **close with finding**, on an audit trail.

376

ranked, cited recommendations across ten cities · draft notice PDF each

live worklist, this morning

PROTECT

Templated advisories (no LLM) in eight scripts — app, Telegram, IVR line, public displays.

Delhi, Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

English · Very Poor, +24 h · templated

Five AI agents on one graph — signal → cited recommendation in **1.0 s** of compute, measured live; approval, dispatch and closure timestamped per action. The spatial unit is the same ~1 km cell everywhere, so every claim traces from a raw reading to an action.

Delhi's winter, replayed — against the government's real orders

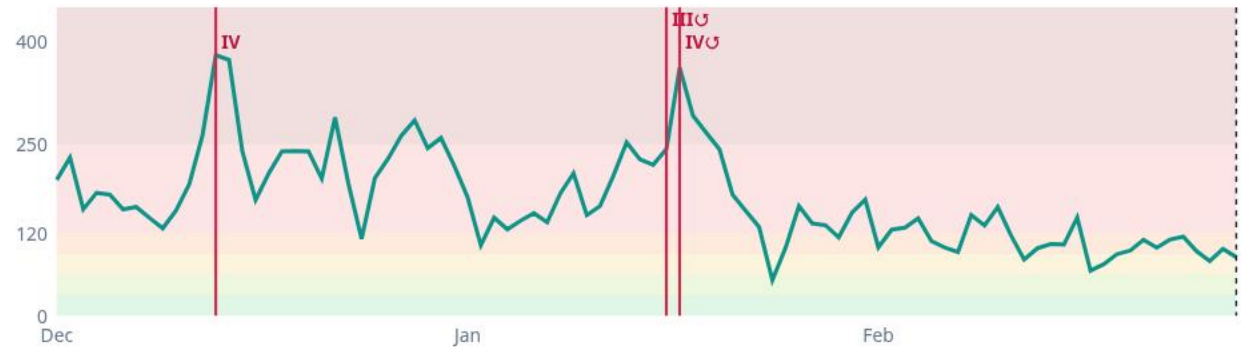
216.3 Dec 2025 mean $\mu\text{g}/\text{m}^3$ · AQI 374 Very Poor **175.3** Jan 2026 mean $\mu\text{g}/\text{m}^3$ · AQI 343 Very Poor **106.6** Feb 2026 mean $\mu\text{g}/\text{m}^3$ · AQI 254 Poor



39 station cells (drawn 2.2x for legibility) · daily mean PM2.5 · CPCB via OpenAQ
outline = Delhi municipal wards; some CPCB stations sit outside it, in the wider NCR

city-mean PM2.5 ($\mu\text{g}/\text{m}^3$) · daily

28 Feb 2026



Stage IV 2025-12-13
24 h before: **P(>120) 83%** · 99% of 38 cells
persistence said 225 · at order 382

Stage III 2026-01-16
24 h before: **P(>120) 85%** · 100% of 1 cells $\triangle$
persistence said 205 · at order 214

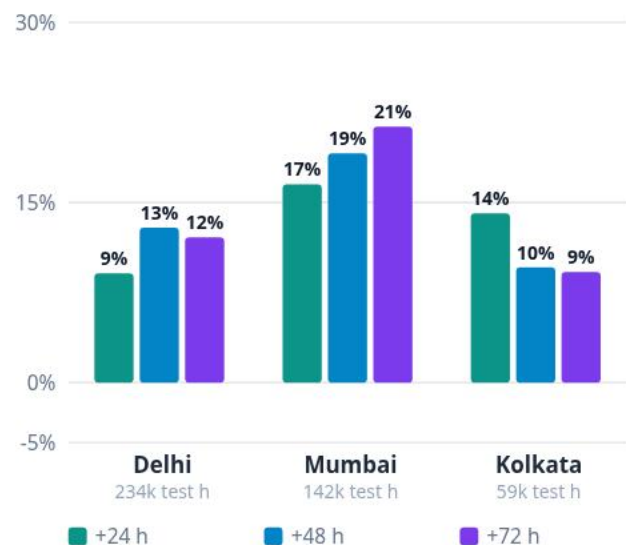
Stage IV 2026-01-17
24 h before: **P(>120) 72%** · 100% of 1 cells $\triangle$
persistence said 131 · at order 333

Served forecast replayed with the production recipe (rolling monthly refit, persistence blend, calibrated P). Green = $P(>120) \geq 30\%$ a day before the order. $\triangle$ = the public feed carried a single Delhi station that week. Sources: CAQM / PIB releases, linked in docs/OUTCOMES.md.

Every number is measured — including our failures

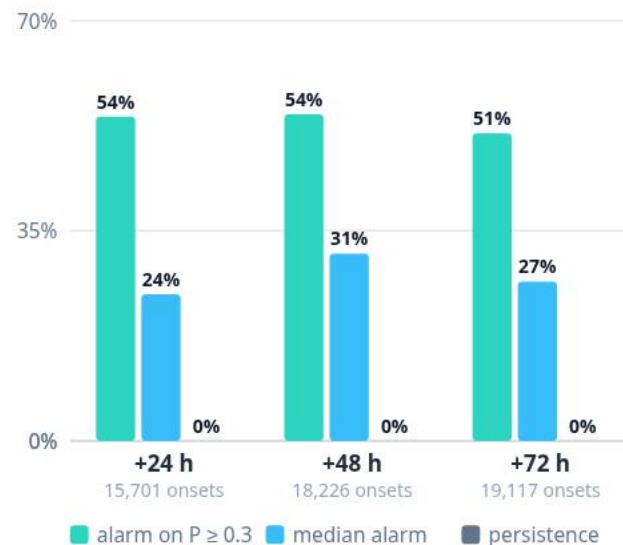
Served forecast skill vs persistence

$1 - \text{RMSE} / \text{RMSE}_{\text{persistence}}$ · strict temporal split, monthly refit on trailing 90 d



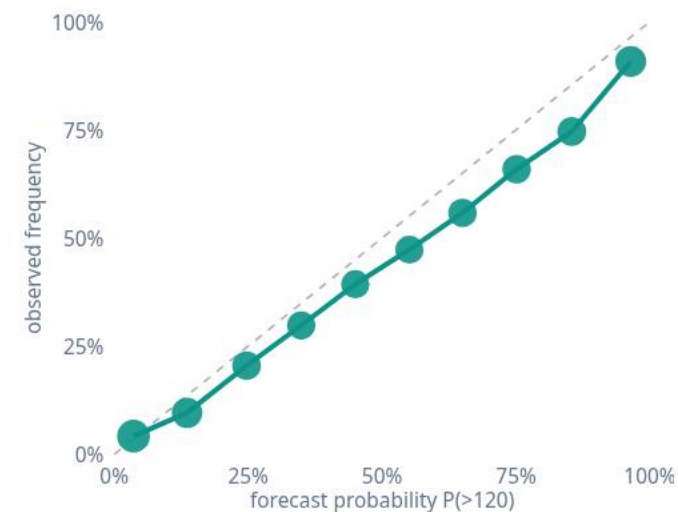
Warns before the air turns

clean → Very-Poor onsets flagged, alarm on calibrated $P \geq 0.3$ · Delhi, multi-season



The probabilities are honest

reliability of $P(>120)$ at +24 h · 80 % band coverage



Attribution vs published apportionment: mean absolute difference 0.042 / 0.099 / 0.097 · bucket tables vs TERI-ARAI 2018 & Guttikunda 2019 — disagreements stated

TFT trained on GPU and rejected — LightGBM won held-out skill

562 backend tests · 17 e2e flows · CI on every push

LIVE DEMO

Delhi, right now.

1

Morning brief → blame map

what changed overnight, where the air turns; click the worst hexagon → sources, SHAP, R² gate, 72-h forecast with P(Very Poor)

2

Worklist → evidence → notice

real Sentinel-2 image + retrieved regulation → draft notice PDF; Approve · Dispatch · **Close case** · History

3

Advisories in eight scripts

Hindi → Tamil; live Telegram bot; IVR line

4

Proof, then ten cities

validation card · real orders in hindsight · scoreboard → Mumbai

press D

opens the live console inside this deck · D returns · offline: the last screenshot is shown, labelled

Built to deploy, not to demo

What it costs to run — measured, not assumed

monthly infrastructure vs cities · one row per reading · 180-day raw retention with archive



Supabase free 500 MB → Pro US\$25; Render free → Starter US\$7 for an always-on API. docs/SCALE.md

RUNNING AT PRODUCTION SCALE — LIVE

10

cities live · Delhi to Lucknow

16,529

~1 km² cells modelled

647

registered + satellite-detected sources

5,495

vulnerability-scored zones

376

RAG-cited enforcement recommendations

8

advisory scripts · app / Telegram / IVR / displays

Adoption path

- **One YAML per city** (bbox, languages, authority) + one backfill run — 7 metros in a week
- **Closed loop:** approve → dispatch → close with finding, immutable audit trail
- **PRANA-ready export** of every tracked action with its measured effect — feeds the official NCAP portal
- Delhi ROI card: **₹0.91 L cr/yr of Delhi's annual health burden avertable at the ~30 % NCAP target (18,219 deaths/yr) — every figure cited**

India measures. India forecasts.

VayuNetra operates.

Traced, predicted, cited, delivered, **closed**. In seconds. In eight scripts. For ₹0.



vayunetra-aqi.vercel.app

github.com/omkarr88/VayuNetra · @aqivayu_bot

Omkar Kadam · Sejal Kumbhar · Abhinav Prasad — thank you.



APPENDIX

For the Q&A

A1 Architecture
A2 Five agents live
A3 Evidence dossier
A4 Citizen reach — eight scripts
A5 Models & validation, all ten cities

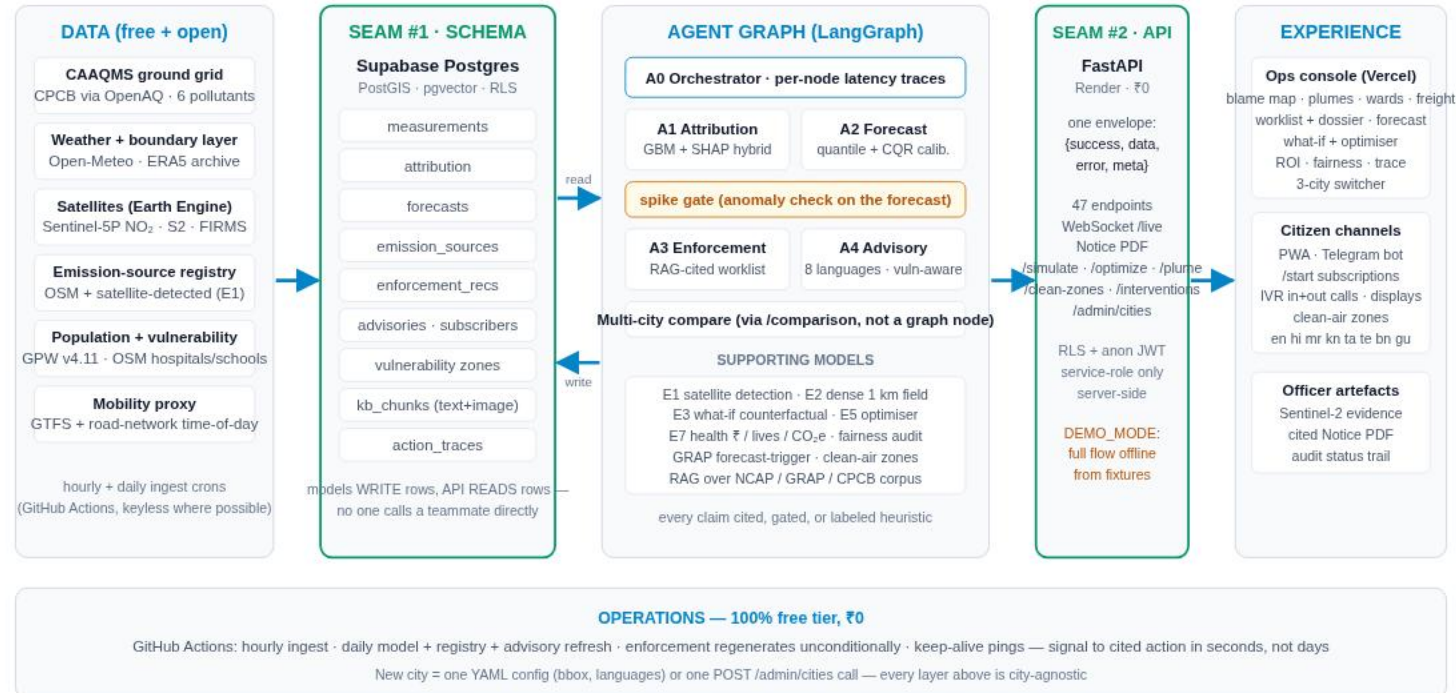
A6 Honesty & fairness
A7 Cost, security & roadmap
A8 Ten cities
A9 Past air
A10 Positioning vs SAFAR / DSS / PRANA

A11 Early warning & calibration
A12 Attribution validation & exposure
A13 Real orders in hindsight — did the air change?
A14 The officer loop closes — audit trail

Signal to cited action — two seams, one envelope

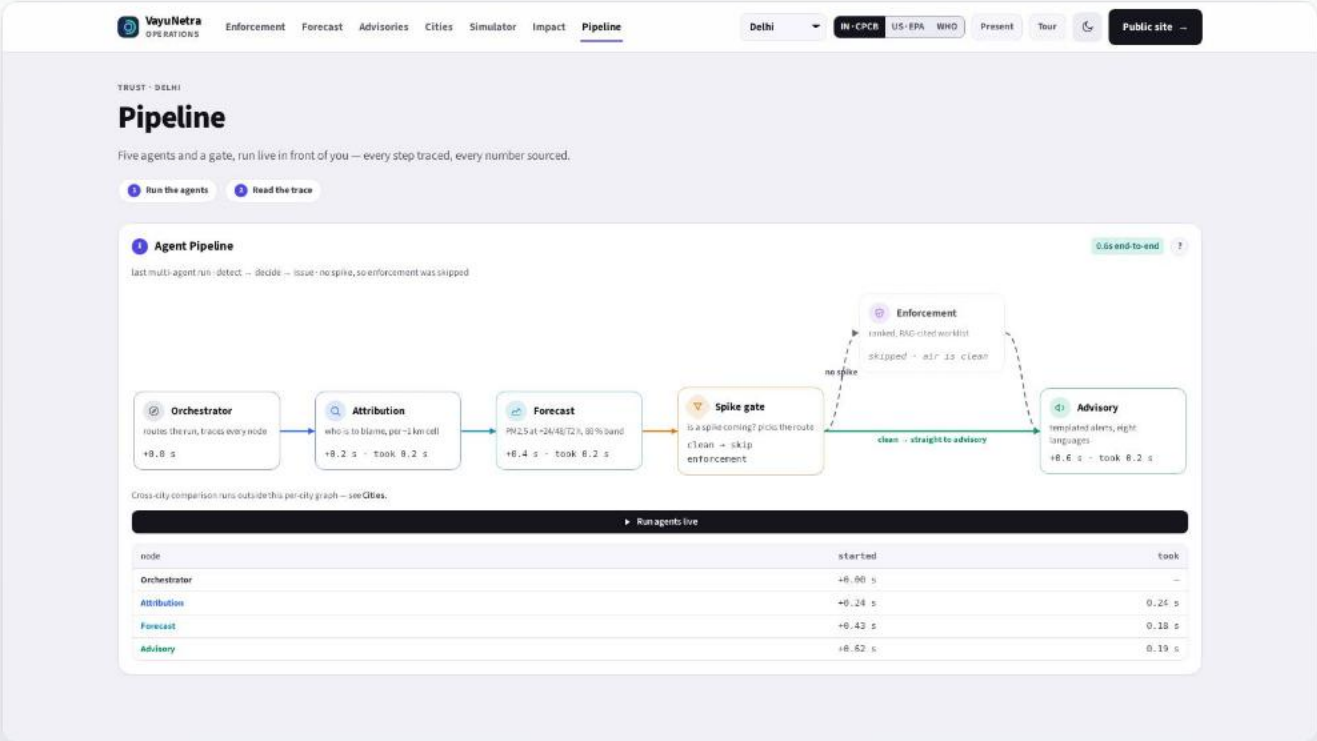
VayuNetra — signal to city action in minutes

Every box decouples through two seams: the Supabase schema and the API contract. Spatial unit everywhere: Uber H3 res-8 (~1 km hexagons).



FastAPI (47 routes + WebSocket) on Render · React + MapLibre / deck.gl on Vercel · Supabase Postgres + PostGIS + pgvector with RLS (writes only through the service role) · LangGraph agents · GitHub Actions crons — all free tier for the ten cities running today.

Five agents on one graph, traced per node



Latest trace, Delhi

attribution agent	351 ms
forecast agent	341 ms
enforcement agent — skipped this run (spike gate: no spike)	-
advisory agent	339 ms
signal → cited recommendation	1.0 s

action_traces · signal → cited recommendation is compute time; approval / dispatch / closure are separate timestamped steps

A recommendation you can defend

The screenshot displays the VayuNetra Operations interface for Delhi. The top navigation bar includes links for Enforcement, Forecast, Advisories, Cities, Simulator, Impact, and Pipeline. The main content area is divided into sections for regulatory citations and construction dust data.

HOW THESE NUMBERS WERE OBTAINED
The share comes from cited chemical-signature priors — this cell's local model did not pass the skill gate (holdout R2 too low for trustworthy apportionment (0.00 < 0.15)).

REGULATORY CITATIONS (RAG)

- Graded Response Action Plan (GRAP)**
Hospitals: additional respiratory care wards activated. Helpline numbers published; mobile health units deployed. match 12%
- National Clean Air Programme (NCAP) — Overview**
daily by CPCB, 2. Performance Review. Quarterly review of city action plan progress by CPCB. Annual ranking of NCAP cities based on PM2.5 reduction. Cities achieving targets: ... match 10%
- Sentinel-2 patch - Concor, Jaipur**
Sentinel-2 RGB median composite (2020-04-18 to 2020-08-18) around Concor, Jaipur (industry) in Jaipur. Detection: Earth Engine heuristic, confidence 0.9. match 9%

Construction dust - Punjabi Bagh
-9 km 16k/hr · 2h · prior-based share · priority 18 · rubric 5/10
Construction dust contributes approximately 79.1% of PM2.5 in this cell (~19,730 residents in this cell, GPM 2020); Satellite-detected construction site #611 is a satellite-detected construction site here (Sentinel-2 heuristic, 60% detection confidence); It is not in any registry. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP); Dust Control Norms For Construction Sites — India.
79% contribution · 19,730 exposed
Evidence dossier Notice PDF Closed - not applicable - 2026-08-18 History

Construction dust - Ward PUSA
-13 km 5k/hr · 2h · prior-based share · priority 3 · rubric 7/10
Construction dust contributes approximately 10.9% of PM2.5 in this cell (~24,632 residents in this cell, GPM 2020); Satellite-detected construction site #76 is a satellite-detected construction site here (Sentinel-2 heuristic, 86% detection confidence); It is not in any registry. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP); Dust Control Norms For Construction Sites — India.
11% contribution · 24,632 exposed
Evidence dossier Notice PDF Approve Dismiss

Construction dust - Ward PUSA
-13 km 3k/hr · 2h · prior-based share · priority 2 · rubric 7/10
Construction dust contributes approximately 10.9% of PM2.5 in this cell (~17,583 residents in this cell, GPM 2020); Satellite-detected construction site #23 is a satellite-detected construction site here (Sentinel-2 heuristic, 81% detection confidence); It is not in any registry. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP); Dust Control Norms For Construction Sites — India.
11% contribution · 17,583 exposed
Evidence dossier Notice PDF Approve Dismiss

Construction dust - Ward PUSA
-13 km 2k/hr · 2h · prior-based share · priority 1 · rubric 6/10
Construction dust contributes approximately 10.9% of PM2.5 in this cell (~7,738 residents in this cell, GPM 2020); Satellite-detected construction site #35 is a satellite-detected construction site here (Sentinel-2 heuristic, 73% detection confidence); It is not in any registry. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP); Dust Control Norms For Construction Sites — India.
11% contribution · 7,738 exposed
Evidence dossier Notice PDF Approve Dismiss

Real Sentinel-2 imagery of the actual site (multimodal RAG over image patches + regulatory text).

Regulation retrieved verbatim — cited chunk by chunk; the right instrument for the city (GRAP only in Delhi-NCR).

Draft notice PDF: addressee, provisions, 24-h deadline, satellite image, projected impact of compliance (screening estimate), provenance block. Stamped **pending officer authorisation**.

The same advisory, in the reader's own script

ENGLISH

Delhi, Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

HINDI

Delhi, Ward 12: अगले 24 घंटों में हवा बहुत खराब रहने का अनुमान है. बाहर की गतिविधि कम रखें, N95 मास्क पहनें, और भारी काम पीक समय के बाद करें.

KANNADA

Delhi, Ward 12: ಮುಂದಿನ 24 ಗಂಟೆಗಳಲ್ಲಿ ಗಾಳಿಯ ಗುಣಮಟ್ಟ ತುಂಬಾ ಕಳಪೆ ಇರಲಿದೆ ಎಂದು ಅಂದಾಜಿಸಲಾಗಿದೆ. ಹೊರಗಿನ ಚಟುವಟಿಕೆ ಕಡಿಮೆ ಮಾಡಿ, N95 ಮಾಸ್ಕ್ ಬಳಸಿ, ಮತ್ತು ಭಾರೀ ಕೆಲಸವನ್ನು ಪೀಕ್ ಸಮಯದ ನಂತರ ಮಾಡಿ.

MARATHI

Delhi, Ward 12: पुढील 24 तासांत हवा खूप खराब राहण्याचा अंदाज आहे. बाहेरील हालचाल कमी ठेवा, N95 मास्क वापरा, आणि जड काम पीक वेळनंतर करा.

TAMIL

Delhi, Ward 12: அடுத்த 24 மணி நேரத்தில் காற்றின் தரம் மிக மோசம் இருக்கும் என எதிர்பார்க்கப்படுகிறது. வெளியில் செல்வதைக் குறைக்கவும், N95 முகக்கவசம் அணியவும், கடின வேலைகளை உச்ச நேரத்திற்குப் பிறகு செய்யவும்.

TELUGU

Delhi, Ward 12: రాబోయే 24 గంటల్లో గాలి నాణ్యత చాలా చెడ్డదిగా ఉండవచ్చు. బయటి కార్యకలాపాలు తగ్గించండి, N95 మాస్క్ ధరించండి, భారీ పనిని పీక్ సమయం తర్వాత చేయండి.

BENGALI

Delhi, Ward 12: আগামী 24 ঘণ্টায় বাতাস খুব খারাপ থাকার পূর্বাভাস। বাইরে কম সময় থাকুন, N95 মাস্ক পরুন এবং ভারী কাজ পিক সময়ের পরে করুন।

GUJARATI

Delhi, Ward 12: આગામી 24 કલાકમાં હવા ખૂબ ખરાબ રહેવાની આગાહી છે. બહારની પ્રવૃત્તિ ઓછી રાખો, N95 માસ્ક પહેરો અને ભારે કામ પીક સમય પછી કરો.

Deterministic templates (no LLM), script-validated; app · Telegram bot · IVR line (Hindi voice for Hindi-first cities) · public displays; 5,495 vulnerability-scored zones (hospitals, schools, elder-care, outdoor work × population).

All ten cities, both baselines — negatives kept

city · window	test hours @24 h	vs persistence 24/48/72 h	vs weekly seasonal-naive	80 % PI coverage
Delhi · multi-season	2,34,491	+9% / +13% / +12%	+26% / +20% / +16%	78% / 78% / 78%
Mumbai · multi-season	1,41,667	+17% / +19% / +21%	+37% / +35% / +32%	82% / 82% / 79%
Kolkata · multi-season	59,373	+14% / +10% / +9%	+26% / +14% / +13%	75% / 73% / 70%
Ahmedabad · live 90 d	854	+15% / +15% / +12%	+17% / +17% / +17%	75% / 83% / 84%
Bengaluru · live 90 d	1,256	+12% / +36% / +41%	+10% / -9% / +1%	83% / 84% / 80%
Chennai · live 90 d	874	+11% / +27% / +25%	+18% / +3% / +2%	81% / 78% / 81%
Delhi · live 90 d	2,407	+12% / +20% / +18%	+31% / +27% / +22%	81% / 75% / 72%
Hyderabad · live 90 d	853	-2% / +6% / +23%	+34% / +25% / +29%	74% / 74% / 72%
Jaipur · live 90 d	1,018	+10% / -1% / -2%	+30% / +15% / +8%	83% / 81% / 65%
Kolkata · live 90 d	556	+14% / +17% / +35%	+3% / +16% / +25%	79% / 80% / 87%
Lucknow · live 90 d	888	+23% / +21% / +36%	+37% / +33% / +26%	74% / 75% / 87%
Mumbai · live 90 d	2,049	+14% / +15% / +5%	+53% / +51% / +29%	73% / 71% / 81%
Pune · live 90 d	1,327	+15% / +8% / +15%	+19% / +15% / +18%	78% / 72% / 68%

Protocol: strict temporal split, rolling monthly refit on the trailing 90 days (= production), one shared support mask; served forecast = LightGBM median blended with persistence, conformal 80 % band, calibrated exceedance probabilities. TFT-lite trained on GPU and rejected. `python -m ml.eval.benchmark · GET /metrics/benchmark`

Honest by construction

The model abstains

No ML blame without out-of-sample skill ($R^2 \geq 0.15$); otherwise cited chemical-signature priors, and the UI says which.

Health text is LLM-free

Deterministic templates in eight scripts — a language model cannot hallucinate medical advice.

Notices are drafts

Stamped pending officer authorisation; never auto-sent. Dispatch starts before/after measurement against the city's drift.

Nothing fabricated

Impact figures return null over invented constants; satellite evidence is real imagery of the real site; study numbers are read from the primary PDFs.

No socio-economic inputs

Priority = contribution × exposure × actionability × confidence. No income or land-value feature anywhere in the pipeline.

Auditable end to end

Per-node agent traces + an immutable status log (who, when, from → to, finding) that outlives the record it describes.

₹0 today — a config file per city, a measured path forward

Stack · ₹0

FastAPI on Render · React + MapLibre on Vercel
· Supabase Postgres (PostGIS + pgvector) ·
LangGraph · GitHub Actions crons. Free tier for
the ten cities running today.

Where ₹0 stops

cities	DB @180 d	₹ / month
10 (today)	~465 MB	0
15	~655 MB	~2,100
50	~2.0 GB	~2,100
131 (all NCAP)	~5.1 GB	~2,700

measured 0.21 MB/city/day; one row per reading; archive-first
retention

Security & roadmap

RLS on every table; public reads via anon JWT;
writes only through the service role on the
server; admin onboarding behind a key header.
Roadmap: InMAP/PAVITRA source-receptor
physics for the what-if engine, a learned
satellite CV detector, more in-language IVR
voices, permit-registry connectors.

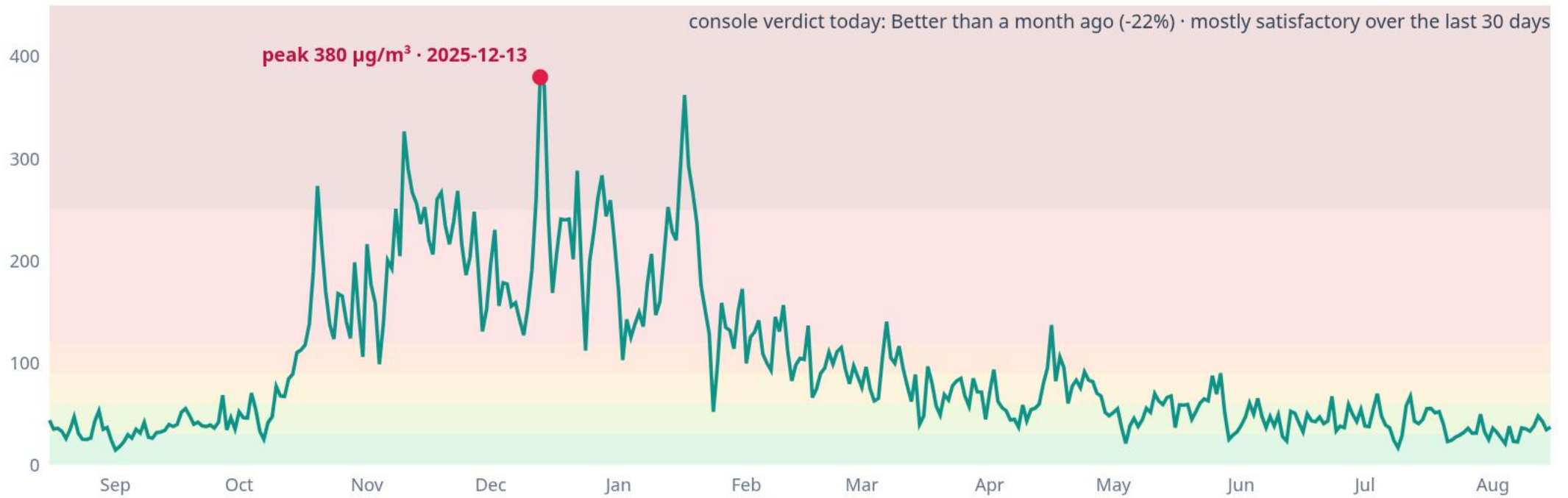
One engine, ten cities — live



#	city	PM2.5 now	+24 h	dominant source	languages
1	Kolkata	40.3	37.2	traffic	bn en hi
2	Jaipur	40.1	37.7	transported	hi en
3	Ahmedabad	30.3	27.4	construction dust	gu en hi
4	Delhi	24.5	34.8	traffic	hi en
5	Lucknow	24.1	17.6	traffic	hi en
6	Bengaluru	21.3	13.2	construction dust	kn en
7	Chennai	19.5	22.0	construction dust	ta en
8	Hyderabad	19.4	18.2	traffic	te en hi
9	Pune	16.7	16.9	construction dust	mr en
10	Mumbai	10.8	11.6	construction dust	mr en hi

live comparison · as of 22 Aug, 10:50 am

Delhi, one year of daily PM2.5 — the real winter



station-day means over 39 Delhi stations (CPCB via OpenAQ) — the same record the forecast benchmark uses; CPCB bands · the console shows the same view from its live table U archived rollout (GET /history/trend)

India can measure and forecast its air. The gap is acting on it — audited.

What exists

SAFAR (IITM, 4 cities, city-level bulletin) · AQEWS / DSS (WRF-Chem, 400 m grid, 29-source — CAQM/government users only) · PRANA (cities report NCAP actions & spend) · NCAP Tracker · CAAQMS / SAMEER (a reading, not a decision).

What the audit found — CEEW, Oct 2025

'Severe and above' caught 5 of 14 episodes (1 of 15 the winter before) · PM2.5 under-predicted by $\sim 24 \mu\text{g}/\text{m}^3$ in winter · CAQM imposed GRAP III/IV on *observed* AQI, not on forecasts · DSS lacks outcome monitoring and actionable short-term pathways.

What VayuNetra adds

Calibrated P(>band) on every cell · onset recall reported vs persistence · live hourly attribution, not a 2016 inventory · forecast → named notice with citations → dispatch → measured outcome, closed with a finding · public, 8 languages, IVR/Telegram/web · ₹0 on the sunk CAAQMS investment.

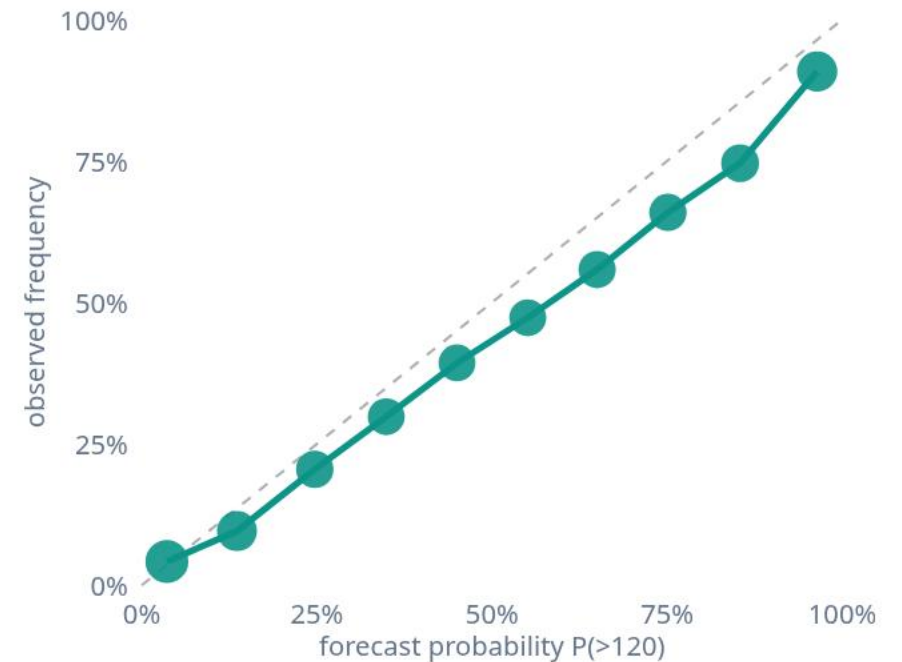
We do not claim to out-model WRF-Chem. Sources: CEEW 'How Well can Delhi Predict Air Quality?' (Ignatious & Rafiuddin, 2025); Ghude et al., GMD 17 (2024); docs/POSITIONING.md

Does it warn before the air turns? Measured on 2025-26 Delhi

horizon	onsets (>120)	onset recall — alarm on $P \geq 0.3$	onset recall — persistence	alarm F1 ($P \geq 0.3$) / persist.	$P(>120)$ Brier skill	80 % PI coverage
+24 h	15,701	54%	0%	0.765 / 0.747	+51%	78%
+48 h	18,226	54%	0%	0.744 / 0.708	+46%	78%
+72 h	19,117	51%	0%	0.725 / 0.696	+38%	78%

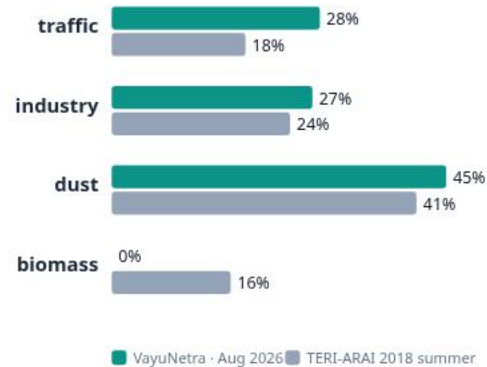
Onset = clean at issue time, above $120 \mu\text{g}/\text{m}^3$ (Very Poor) at $t+h$. 'Tomorrow = today' has onset recall 0 by construction. **Where we are weak** — the Severe tail (>250): the blended forecast only matches persistence there and the median alarm sees few Severe onsets (as the official WRF-Chem does); the product speaks in calibrated probability: $P(>250)$ Brier skill +31 % at 24 h.

Reliability, $P(>120)$, +24 h

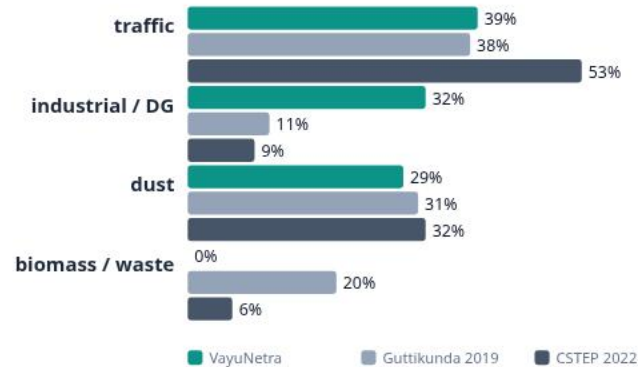


Checked against published apportionment — agreements and disagreements both shown

Delhi · source-only shares



Bengaluru · source-only shares



Who is in the forecast?

+24 h · Very Poor **0.1% · ~26,846 people**

+48 h · Very Poor **0.1% · ~13,331 people**

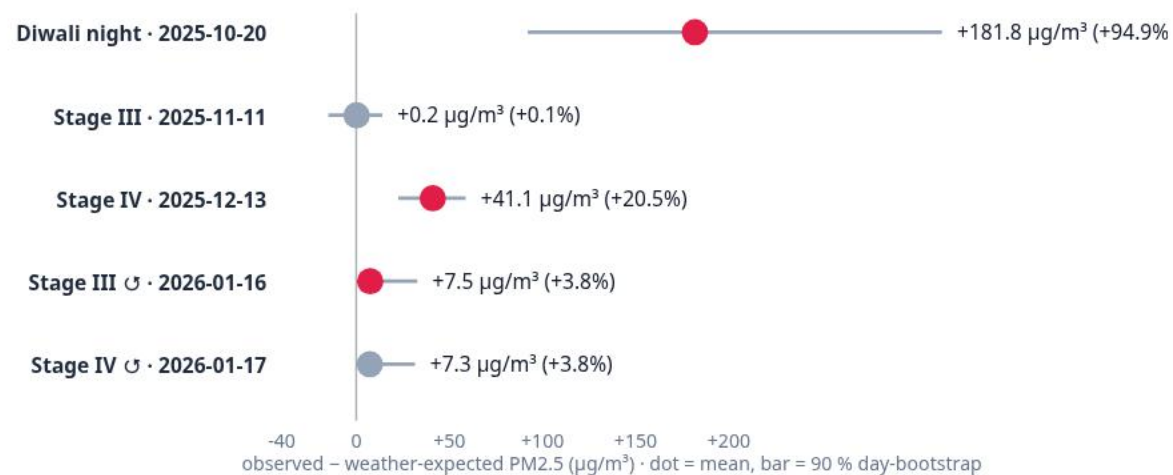
+72 h · Very Poor **0.1% · ~21,985 people**

basis: uniform_city_population (2,06,00,000 people in sampled cells; city-scaled figure assumes the same share city-wide — the card says so)

Σ cell population × calibrated $P(> \text{band})$; GPW v4.11 where sampled, cited city population otherwise. Exposure, not mortality. GET /exposure

Read honestly: rankings agree; we over-read kerbside traffic (arterial-road stations, NO₂/CO signatures) and book less as regional than a dispersion model; our Bengaluru 'industrial' is DG sets, our biomass misses small waste fires. Shares move between daily runs where NO₂ coverage is thin — shown, not smoothed. docs/ATTRIBUTION_VALIDATION.md

Did the air change during the order, weather taken out?



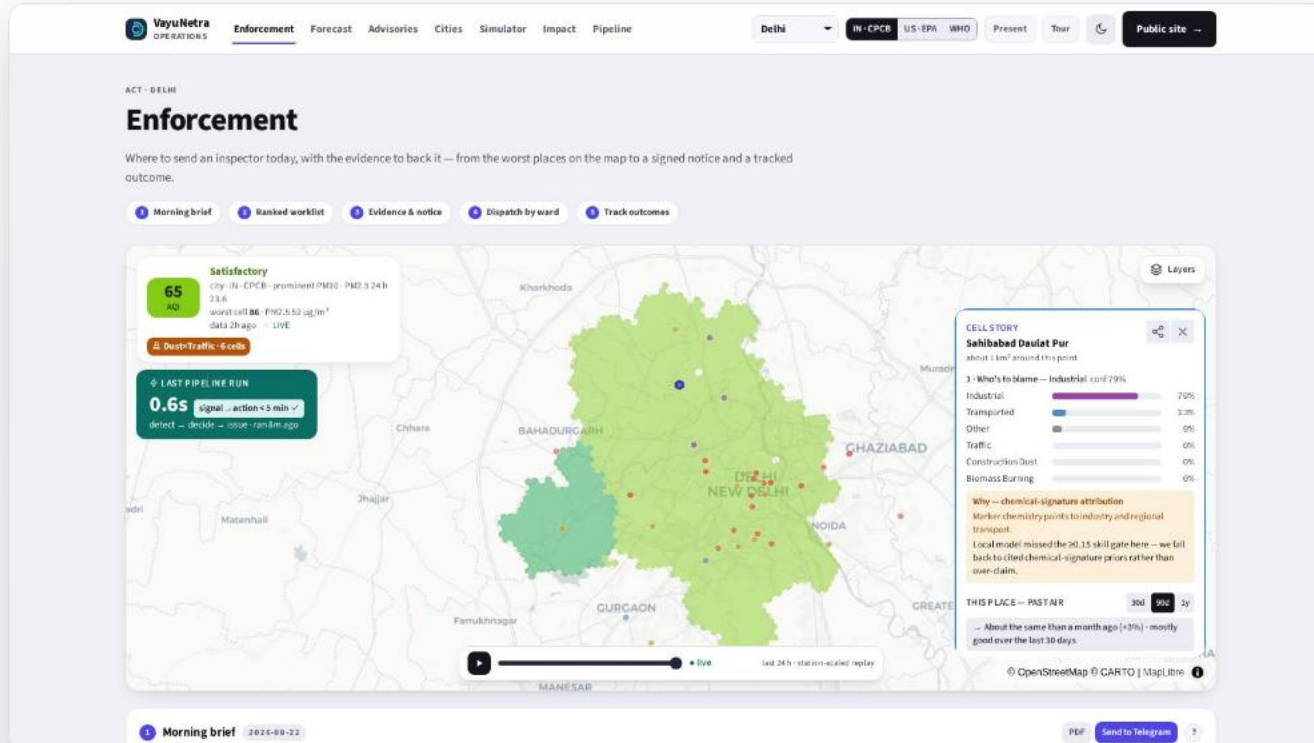
LightGBM on ERA5 meteorology + hour/dow/doy/cell, fitted on 88,270 season hours outside the windows; held-out R^2 0.61, RMSE 65 µg/m³. python -m ml.eval.interventions

Read honestly

Diwali night shows the method has the power to see a large signal. For the GRAP windows we find **no weather-adjusted reduction this method can detect**; the Stage IV rows sit in the most stagnant weather of the season, where a tree model fitted on calmer hours under-predicts. Association, not causation — coincident factors stay in the number.

This is why VayuNetra measures each dispatched action against **its own cell and the city's drift** — a targeted action has a control; a blanket stage does not.

See → decide → act → prove — with an audit trail



1 **Approve** — moves the item to its ward queue

2 **Dispatch** — freezes the cell's 7-day baseline; effect vs city drift is measured from then on

3 **Close with a finding** — violation found · compliant · inaccessible · n/a, plus a note

4 **History** — immutable log: from → to, who, when, finding; survives the nightly run and even deletion of the record

PRANA-ready CSV — every tracked action with baseline / after / effect / status / closure, mapped to its NCAP head.